Project Plan

Waveform Tuning Via Predictive Modelling

Canon Production Printing / Fontys HBO-i Project 4

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1 Introduction and Context

1.1 Company Background and Educational Context

Canon Production Printing is a Netherlands-based subset of Canon headquartered in Venlo that develops, manufactures, and sells printing and copying hardware and related software. Its product range includes office printing and copying machinery, production printers, and wide-format printers for both technical documentation and colour-display graphics.
The project is carried out in the context of Fontys University of Applied Sciences, HBO-i (higher professional education ICT) Project 4, at autonomy and complexity Level 2.

1.2 Project Context

The main topic of this project is the calibration and optimisation of printheads via waveform tuning. Waveform tuning is currently a highly manual and iterative process. It involves adjusting electrical parameters (e.g., voltage, timing, waveform shape) per nozzle to achieve consistent print quality. This process does not scale well due to large parameter space,variability across nozzles, and a lack of systematic, data-driven understanding.Poor tuning leads to defects such as inconsistent droplet formation, trajectory errors, and visible print artifacts.

1.3 Project Goal

The main goal of this project is to develop a data-driven approach to understand relationships between waveform parameters and colour output, build predictive models that estimate color outcomes, and explore the feasibility of suggesting waveform configurations that improve macro-uniformity across printheads.

2 Project Goals and Expected Results

2.1 Scope

The project comprises both technical and organisational dimensions. On a technical scope, it includes:

- Exploratory analysis of the provided waveform colour dataset
- Modelling, evaluation, and reporting of the data
- Development of baseline predictive models and its evaluation
- Alignment with the company's expectations: business narrative, visualisations, process description

On the organisational dimension, it includes:

- Task tracking via a shared backlog.
- Version control using Git with defined branching and commit conventions
- Peer review process for code, analysis, and documentation
- Regular team communication
- Stakeholder communication with Canon (progress updates, feedback sessions)
- Documentation management (consistent naming, structure, and versioning)
- Knowledge sharing within the team (data understanding, findings, decisions)

The project does not include:

- Hardware modifications or physical experiments
- Full-scale deployment within Canon systems
- Guaranteeing optimal real-world performance without validation
- Processing the entire dataset if computationally infeasible

2.2 Data and Technical Context

- Dataset size: up to 725 million samples
- Parameters include
  - Voltage (V)
  - Flank ratio (Fr)

  - Timing variables (e.g., dt2)
  - Coverage levels
  - Multiple colors, chips, and nozzles

Tools and Technologies:

- Python (primary)
- Optional: R
- Git for version control
- Notebooks for analysis and reproducibility

2.3 Intermediate Results and Deliverables

Canon may receive intermediate results through a channel to be agreed (e.g. email, shared folder, or meeting walkthrough). These may include:

Deliverable	Description
Project plan	Scope, organisation, planning, risks
Process & data description	Current tuning process; data structure and dictionary
Exploratory analysis	Summary statistics, visualisations, hypotheses
Cleaned/analysis-ready datasets	Documented subsets or pipelines
Predictive models & evaluation	Metrics, comparisons, limitations
Final report / presentation	Business case, results, recommendations
Repository artefacts	Code, notebooks, documentation (Git)

- Cleaned and documented dataset.
- Exploratory data analysis report (relationships, distributions, anomalies).
- Baseline model(s) with evaluation metrics.
- Short tuning guidance summary.
- Final project report and presentation material.

2.4 Desired Final Result

The desired outcomes include:

- Understanding of how waveform parameters (e.g. voltage, flank ratio, timing) relate to colour outcomes in the available data.
- A predictive model (or family of models) that can suggest or rank waveform configurations that improve uniformity and consistency across printheads/nozzles, within the limits of the data.

- Documentation of the approach (business understanding, data, modelling choices, evaluation), aligned with CRISP-DM, so that Canon can assess reproducibility and next steps (e.g. deployment or further experiments).

Canon indicated no rigid product specification; valuable deliverables include a business case, visualisations, and clarity on the current process versus the proposed data-driven process.

3 Approach

3.1 Project Activities

Activities follow **CRISP-DM** (Cross-Industry Standard Process for Data Mining):

- **Business understanding** — Clarify goals with Canon; map current waveform-tuning process; define success criteria for models and visualisations.
- **Data understanding** — Explore scale and structure of the dataset (waveform sets, chips, nozzles, colours, coverage levels); document variables and quality issues.
- **Data preparation** — Cleaning, subsetting (if needed for feasibility), feature engineering, train/validation splits appropriate for the problem.
- **Modelling** — Select and train predictive models linking waveform configuration to colour (and/or derived metrics); consider physical/mathematical intuition where relevant.
- **Evaluation** — Validate with held-out data and domain-relevant metrics; compare baselines; discuss limitations.
- **Deployment (as study outcome)** — Recommendations for integration or further experiments rather than full production deployment within one semester, unless scope explicitly expands.

Supporting activities: literature and domain research, regular sprint planning and retrospectives, maintaining documentation and reproducible notebooks/scripts, and communication with Canon (including sharing intermediate results in an agreed form).

3.2 Quality Control

- **Version control**: traceable changes, peer review via pull requests where possible; commit and branch naming per `documentation/naming-conventions/git-branch-commit-conventions.md`.
- **Document naming**: new markdown and snapshots follow `documentation/naming-conventions/document-naming-conventions.md` (kebab-case, ISO dates for dated notes).
- **Reproducibility**: fixed random seeds where relevant; documented environment (e.g. Python version, key libraries).
- **Validation**: train/validation/test discipline; avoid leakage across chips or experimental units where applicable; report uncertainty and failure cases.
- **Code quality**: readable structure, naming aligned with data dictionary; optional linting/tests for critical pipelines.

- **Stakeholder feedback**: regular alignment with Canon on direction and interpretation of results.
- **Academic standards**: honest reporting of limitations; citations for methods and domain concepts.

3.3 Team Members

Roles and task ownership are distributed per sprint; the PDP links learning outcomes to activities (e.g. software analysis, design, realisation for some members).

3.4 Collaboration

- **Agile**: sprints, stand-ups or async updates, backlog refinement, retrospectives.
- **GitHub**: shared repository, branches for features/docs, CI checks where configured (e.g. commit/PR conventions).
- **Canon**: contact person uses Python; expectations include business case, visualisation, and process documentation; communication channels to be kept reliable (early project risk if email response is slow—see risk analysis).

Planning is iterative; indicative phases (semester-length) are:

Phase	**Time**	**Focus / ownership**
Kick-off & CRISP-DM	Weeks 1–3	Whole team: Canon alignment, data inventory, EDA start
Data prep & modelling start	Weeks 4–8	Split by strength: data pipeline, EDA, first models
Modelling & evaluation	Weeks 9–12	Model iteration, validation, visualisations
Consolidation	Weeks 13+	Final report, business case, presentation, handover

4 Costs and Benefits

4.1 Costs

- **Student time**: main "cost" — four FTE-equivalent student effort over one semester.
- **Infrastructure**: minimal if using university or free-tier compute; possible cloud costs if large-scale training is needed (to be agreed and kept bounded).
- **Canon time**: stakeholder meetings and feedback.

4.2 Benefits

- For **Canon**: structured insight into waveform - colour relationships; foundation for reducing manual tuning; visual and narrative material for internal decisions; potential path toward tooling.
- For **students**: level-2 project experience, CRISP-DM practice, domain exposure to industrial inkjet/waveform problems, and professional skills development.

5 Risk Analysis

Risk	**Impact**	**Mitigation**
Poor communication with Canon	Wrong priorities, delayed feedback	Agree cadence and channel; escalate early; document decisions
Dataset too large / messy	Slipped timeline	Sample, aggregate, phased deliverables; clear data-quality report
Model underperforms	Weak predictive value	Strong baselines, uncertainty quantification, recommend further data collection
Scope creep	Incomplete final deliverables	Strict backlog; refer to section "Project scope"
Team availability imbalance	Bottlenecks	Transparent task board; pair work; sprint commitments
Technical debt	Hard to reproduce results	Reviews, minimal reproducible pipeline early

1 Appendix: References in Repository

Technical terms (waveform, PZT, banding, IQQ, etc.) are documented in:
`documentation/data-dictionary.md`
`documentation/technical-glossary.md`
Dataset structure and meeting notes:
`documentation/data-set-description.md`,
`documentation/meeting-notes/canon/meeting-notes-canon-2026-03-05.md`.
Naming rules for files and Git: `documentation/naming-conventions/`.

2 Appendix: Methodology Reference

CRISP-DM: Wirth, R. & Hipp, J., “CRISP-DM: Towards a standard process model for data mining,” Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining, 2000.